

## Multiple Inheritance Practice Questions (15 Questions)

These questions focus on:

- Multiple Inheritance
- Taking attributes from multiple parents
- Using super()
- Calling parent class constructors directly
- Displaying all details

Eg :

```
class Father:
```

```
    def __init__(self, father_name, **kwargs):  
        self.father_name = father_name  
        super().__init__(**kwargs)
```

```
class Mother:
```

```
    def __init__(self, mother_name, **kwargs):  
        self.mother_name = mother_name  
        super().__init__(**kwargs)
```

```
class Child(Father, Mother):
```

```
    def __init__(self, father_name, mother_name, child_name):  
        super().__init__(  
            father_name=father_name,  
            mother_name=mother_name  
        )  
        self.child_name = child_name
```

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### 1. Father + Mother → Child (Easy)

**Father**

- father\_name
- father\_age

#### **Mother**

- mother\_name
- mother\_age

#### **Child**

- child\_name

Display all details.

**Requirement:** Call parent constructors directly.

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### **2. Teacher + Engineer → Student**

#### **Teacher**

- teaching\_subject

#### **Engineer**

- programming\_language

#### **Student**

- student\_name

Display all details.

**Requirement:** Use parent class names.

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### **3. Mobile + Laptop → Electronics**

#### **Mobile**

- mobile\_brand

#### **Laptop**

- laptop\_brand

#### **Electronics**

- owner\_name

Display all details.

**Requirement:** Use super() if possible.

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#### 4. CricketPlayer + FootballPlayer → Athlete

##### CricketPlayer

- cricket\_team

##### FootballPlayer

- football\_team

##### Athlete

- player\_name

Display all details.

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#### 5. BankAccount + Loan → Customer

##### BankAccount

- account\_number

##### Loan

- loan\_amount

##### Customer

- customer\_name

Display complete customer details.

---

#### 6. Vehicle + Insurance → Car

##### Vehicle

- vehicle\_number

##### Insurance

- policy\_number

##### Car

- owner\_name

Display all details.

---

## 7. Employee + Trainer → DataScientist

### Employee

- employee\_id

### Trainer

- batch\_name

### DataScientist

- employee\_name

Display all information.

---

## 8. School + Hostel → Student

### School

- school\_name

### Hostel

- room\_number

### Student

- student\_name

Display all details.

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## 9. YouTubeCreator + Blogger → Influencer

### YouTubeCreator

- subscribers

### Blogger

- blog\_name

### Influencer

- influencer\_name

Display all information.

---

## 10. Doctor + Researcher → Scientist

### Doctor

- specialization

### Researcher

- research\_area

### Scientist

- scientist\_name

Display all details.

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## Multiple Inheritance + Abstraction

## 11. Bank + Insurance → Customer

### Abstract Class

Bank

Abstract Method:

- show\_bank\_details()

Attribute:

- bank\_name

### Insurance

Attribute:

- policy\_name

### Customer

Attribute:

- customer\_name

Implement abstract method and display all details.

---

## 12. Animal + Pet → Dog

### Abstract Class

Animal

Attribute:

- animal\_name

Abstract Method:

- make\_sound()

### **Pet**

Attribute:

- owner\_name

### **Dog**

Display all information.

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## **13. Employee + Department → Manager**

### **Abstract Class**

Employee

Attribute:

- employee\_name

Abstract Method:

- work()

### **Department**

Attribute:

- department\_name

### **Manager**

Implement work() and display all details.

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## **14. Course + Institute → Student**

### **Abstract Class**

Course

Attribute:

- course\_name

Abstract Method:

- duration()

### **Institute**

Attribute:

- institute\_name

### **Student**

Attribute:

- student\_name

Display all details.

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## **15. Vehicle + FuelSystem → Car**

### **Abstract Class**

Vehicle

Attribute:

- brand

Abstract Method:

- start()

### **FuelSystem**

Attribute:

- fuel\_type

### **Car**

Attribute:

- model

Display all details.